Deployment & Cost Plan

Clinical Intelligence Agent — Deployment & Cost Plan
Approach: **Amazon EKS (Kubernetes)**
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Currency: USD (approximate; not an AWS quote)

Disclaimer: Costs are rough estimates for planning. Actual AWS, LLM, and Deepgram bills depend on region, usage, and configuration. Always set AWS Budgets and billing alerts.

1. Project Overview

The Clinical Intelligence Agent is a doctor-facing system that transcribes clinical conversations, labels doctor/patient roles, summarizes encounters, extracts entities, retrieves NICE guidelines via RAG, ranks differential diagnoses, and generates analysis + SOAP PDFs.

Current application tech stack (shared by both deployment approaches)

Layer	Technology	Role
API	FastAPI + Uvicorn	REST endpoints
Agent	LangGraph + MemorySaver	Stateful clinical pipeline
LLM	OpenRouter (e.g. DeepSeek) / optional Gemini/Groq	Summary, entities, diagnosis, insights
ASR	Deepgram Nova	Audio transcription + diarization
RAG	FAISS + HuggingFace MiniLM + metadata routing	Guideline retrieval
Observability	LangSmith	Tracing / monitoring
Reports	ReportLab PDFs	Analysis + SOAP notes
Frontend (planned)	Separate SPA (React/Next or similar)	Doctor UI
Auth (planned)	JWT / doctor user accounts	Session isolation

3. EKS Architecture

- One EKS cluster with separate namespaces: frontend, backend
- Frontend Deployment (nginx serving SPA) OR preferred: S3+CloudFront still for UI cost savings
- Backend Deployment (FastAPI) + Service + Ingress (ALB via AWS Load Balancer Controller)
- Independent Helm charts / GitOps (Argo CD) for FE vs BE rollouts
- HPA for each Deployment; Cluster Autoscaler or Karpenter for nodes
- RDS Postgres + S3 + Secrets via External Secrets / IRSA
- Later: backend-api vs backend-worker Deployments + SQS
- Note: EKS control plane is billed even when workloads are idle (~$0.10/hour)

3.1 AWS / K8s tech stack for this approach

Component	Technology	Purpose
Orchestration	Amazon EKS	Managed Kubernetes control plane
Nodes	Managed node groups / Karpenter	Run pods; scale capacity
Frontend	K8s Deployment or S3+CloudFront	Doctor UI

Component	Technology	Purpose
Backend	K8s Deployment + Service	FastAPI clinical API
Ingress / TLS	AWS LB Controller + ALB + ACM	app.* / api.* HTTPS
Scaling	HPA + Cluster Autoscaler/Karpenter	Pod and node autoscaling
Registry	ECR	Container images
GitOps (optional)	Argo CD / Flux	Declarative separate deploys
Data	RDS + S3	Users/visits + files/index
CI/CD	GitHub Actions → ECR → Helm/Argo	Independent FE/BE pipelines

4. AWS Monthly Cost Estimates (EKS)

AWS item	Staging (minimal)	Light prod (HA)
EKS control plane	~$72	~$72
Worker nodes (2 small+)	$30–70	$60–120
ALB	$16–25	$16–25
S3 + CloudFront (frontend)	$1–5	$3–15
ECR	~$1	$1–3
RDS Postgres	$0–15	$15–40
NAT Gateway (typical)	$30–35	$30–35+
CloudWatch / addons	$5–15	$10–25
AWS subtotal (typical)	**$150–230**	**$200–330**

Even an almost-empty EKS cluster still incurs the control plane charge (~$72/month).

2. External API Costs (LLM + Deepgram + LangSmith)

These costs apply to **both** EKS and ECS deployments. They often exceed AWS compute for AI apps.

2.1 Deepgram (Speech-to-Text)

Usage scenario	Assumptions	Rough monthly cost
Light demo	~50 consultations × 5 min = 250 min	~$5–15
Small clinic	~300 consults × 8 min = 2,400 min	~$30–80
Growing usage	~1,000 consults × 10 min = 10,000 min	~$100–300+

Note: Exact Deepgram pricing depends on model (Nova), features (diarization), and plan. Check current Deepgram pricing page.

2.2 LLM via OpenRouter / model APIs

Each analyze run may call the LLM multiple times: role labeling, summary, entities, RAG diagnosis, insights/SOAP.

Scenario	Assumptions	Rough monthly cost
Light demo	~50 full analyses; cheaper model (DeepSeek-class)	~$5–25
Small clinic	~300 analyses; mixed prompts	~$40–120
Heavier / stronger models	~300 analyses; higher-tier models	~$100–300+

Cost drivers: tokens in/out, number of LLM calls per visit, model choice, follow-up loops, PDF insight generation.

2.3 LangSmith

- Free/developer tiers may cover early demos
- Production tracing volume can add ~$0–50+/mo depending on plan and sampling
- Recommendation: sample traces in production (e.g. 10–20%) to control cost

2.4 Combined external API ballpark

Stage	Deepgram	LLM	LangSmith	API total
Demo	$5–15	$5–25	$0–10	**$10–50**
Small clinic	$30–80	$40–120	$0–30	**$70–230**
Growth	$100–300	$100–300	$20–50	**$220–650+**

5. Total Cost of Ownership (AWS + APIs)

Stage	AWS (EKS)	External APIs	Combined rough total
Demo / staging	$150–230	$10–50	**$160–280 / month**
Small clinic prod	$200–330	$70–230	**$270–560 / month**
Growth	$300–450	$220–650	**$520–1,100+ / month**

6. Scaling & Cost Optimization on EKS

- Prefer S3/CloudFront for frontend even in an EKS org — saves node capacity
- Separate node pools: small for frontend/addons, larger memory nodes for backend/RAG
- Use Karpenter + Spot for workers/staging (careful on prod API)
- HPA on Deployments; scale workers to 0 when queue empty
- Avoid multi-AZ NAT until required (NAT is expensive)
- Same LLM/Deepgram optimizations as ECS approach
- EKS is justified when Kubernetes is an org standard or you run many services/teams

7. Delivery Phases

- P0: EKS cluster + ALB controller + metrics-server
- P1: Backend Deployment/Service/Ingress
- P2: Frontend deploy (K8s or S3) + separate pipeline
- P3: HPA + Cluster Autoscaler/Karpenter
- P4: RDS + S3 + replace in-memory state for multi-pod safety
- P5: GitOps (Argo) staging/prod + budgets
- P6: API/worker split for analyze workloads

8. Comparison Snapshot vs ECS Fargate

Topic	EKS	ECS Fargate
Staging AWS	~$150–230	~$45–90
Light prod AWS	~$200–330	~$90–180
Idle cost	Control plane always ~$72	Pay mainly for running tasks/ALB

Topic	EKS	ECS Fargate
Ops complexity	Higher	Lower
Best for this project	If K8s is required	**Recommended default**

Note: Free Tier cannot cover a real EKS setup. Use local kind/minikube to practice manifests at $0, then budget for paid AWS when deploying EKS.

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